

MUHAMMAD HASIF BIN MOHD FAISAL

UNDERGRADUATE STUDENT

Singapore 400412 ♦ +6588966503 ♦ muhammad.hasif.faisal@gmail.com

Analytically driven and technically versatile individual with hands-on experience across data analysis, cybersecurity, automation, and operational technology. Proven track record supporting national-level initiatives through real-time data systems, geospatial dashboards, RPA solutions, and predictive modeling. Adept in Python, SQL, ArcGIS, and BI tools, with a deep understanding of stakeholder needs, cross-agency collaboration, and applied machine learning. Known for leading teams under pressure, adapting quickly in dynamic environments, and communicating complex ideas with clarity. Committed to delivering secure, scalable, and data-informed solutions across sectors.

EDUCATION

Bachelor of Science with Honors: Applied Artificial Intelligence	Expected in 02/2028
Singapore Institute of Technology - Singapore	
Diploma of Education: Financial Business Informatics	04/2023
Temasek Polytechnic - Singapore	
• Major Project, Time-Series Forecasting Techniques - Research Project	
GCE 'O' Levels	04/2019
Al-Arabiah Institution - Singapore	
• Grade: Distinction in Mathematics	
• Combination: Triple pure Science + A Math	
PSLE	12/2015
Eunos Primary School - Singapore	
• Grade: PSLE T Score of 235	

EXPERIENCE

Research Engineer/ Developer (Student), (Full-Time)	09/2024 - 12/2025
Singapore Institute of Technology (SIT)	
Worked within an agile research-oriented development team to enhance and extend R3CAP, an open-source platform for near real-time 3D reconstruction and collaborative scene authoring (previously known as demoConstruct). This platform enables accessible creation and editing of digital twin environments, integrating progressive 3D reconstruction, WebXR, real-time rendering, and cloud services.	
Key Projects & Contributions:	
• Feature Enhancements & Integration Led implementation of major front-end and back-end features using Python and Babylon.js, improving user workflows such as scene editing, object deletion, and real-time mesh replacement within the collaborative 3D environment.	
• 3D Reconstruction Engine Integration Integrated next-generation real-time 3D reconstruction and segmentation engines (e.g., progressive reconstruction and auto-labeling) into the platform, enabling dynamic model updates and real-time segmentation for enhanced digital twin creation.	
• Research-Driven Development Contributed to research outputs and academic submissions, supporting findings for publications at CHI '25 and DIS '26 through prototype enhancements, experimental evaluations, and documentation of results.	
Section Leader, Operations Department, (Full-Time)	06/2024 - 08/2025
Singapore Police Force (SPF) - National Service	
Completed officer training and supported national-level operations through data-driven planning, geospatial analytics, and workflow automation, working across SPF, HTX, and GovTech on high-security and large-scale public events.	
Key Projects & Contributions:	
• Movement Tracker (HTX Collaboration) Co-developed a personnel movement monitoring system for high-security operations using geospatial data to improve real-time visibility and coordination.	
• Geospatial Operations Dashboards (ArcGIS) Designed and deployed interactive dashboards and map layers to visualize deployments and enhance situational awareness.	
• Polling Operations Automation (RPA Integration) Developed RPA scripts to automate backend polling logistics workflows for GE2025, reducing manual effort and operational overhead.	
• VoteQ Platform Enhancement (GovTech Collaboration) Integrated ArcGIS-based spatial analytics into voteQ to improve crowd monitoring and resource allocation.	
• Vice-Chairperson, Cadet Councils Led cohort initiatives, coordinated training activities, and represented cadet interests to command staff.	
• 19-Day Outward Bound Singapore Expedition Selected for and completed immersive leadership and resilience training focused on navigation, teamwork, and adaptive problem-solving.	
• Awarded Best Operational Fitness Recognizing best in performance under sustained physical and operational demands.	

Core Tasks

- Completed the Senior Officer Course (80th Batch), covering leadership, operational execution, organizational management, and strategic planning.
- Led squad-level training exercises, applying structured decision-making and communication under time-critical conditions.
- Designed dynamic Excel-based planning tools for manpower tracking, scenario modeling, and resource allocation.
- Conducted data validation, cleansing, and reconciliation to ensure accuracy of operational plans.
- Led technical onboarding and authored user guides to standardize ArcGIS and RPA adoption across SPF units.
- Managed operational scheduling, clearance workflows, and logistical coordination for deployments.
- Provided on-site systems and data support during live exercises and national events, including National Day Parade (NDP) and nationwide readiness drills.

Cyber Operations Engineer | HTSOC (Full Time)

01/2024 - 06/2024

HTX (Home Team Science & Technology Agency) / Singapore Police Force (SPF) - National Service

Completed foundational police training and was handpicked as a full-time NSF to join HTX's Cyber Operations team, supporting national cybersecurity operations within a Security Operations Center (SOC).

Key Projects & Contributions

- **Cyber Operations Engineer, HTSOC**
Selected to operate within HTX's SOC, supporting digital defense, incident response, and high-profile national engagements. SOC
- **Automation & Reporting**
Developed scripts to automate SOC reports and dashboard generation, improving operational visibility and accelerating insights delivery.
- **Incident Response & Digital Forensics**
Conducted forensic investigations and coordinated incident-response playbooks to support rapid threat mitigation and evidence preservation.
- **Proactive Threat Hunting**
Executed cyber threat hunts to identify emerging vulnerabilities and remediate internal threats, including phishing campaigns. **High-Profile**
- **Event Support**
Supported cybersecurity demonstrations and briefings during the HTX TechX Summit and ministerial visits, translating complex technical concepts for diverse audiences.
- **Squad In-Charge (POBC)**
Led and managed a squad during basic training, overseeing drills, welfare, and administrative coordination.
- **Awards & Selection**
Awarded Best Operational Fitness and Best Trainee, recognizing top performance and discipline.
Selected for the Officer Cadet Course, reflecting strong technical competence and leadership potential.

Core Tasks

- Conducted continuous security monitoring, threat analysis, and incident triage within a SOC environment.
- Enforced compliance standards and maintained security documentation, including post-incident reports.
- Collaborated with cross-functional teams across HTX, SPF units, and GovTech on coordinated response efforts.
- Managed privilege-protected systems in accordance with HTX security protocols, ensuring data confidentiality.
- Completed the Police Operational Basic Course, covering physical conditioning, tactics, and operational fundamentals.

Workforce Data Insights Officer , (Full Time)

04/2023 - 07/2023

Ministry of Manpower

Supported workforce development initiatives by designing and executing surveys, interviews, and outreach campaigns to gather actionable labor-market intelligence under government protocols.

Key Project

- **Workforce Survey Framework**
Developed and rolled out a structured survey and interview protocol to capture employment trends and stakeholder perspectives, ensuring robust data quality and compliance.

Core Tasks

- Designed and administered surveys and structured interviews in line with regulatory guidelines to inform employment policy.
- Conducted outbound outreach via phone and email to gather stakeholder input and clarify responses.
- Collected, validated, and entered quantitative and qualitative data, maintaining adherence to government standards.
- Collaborated with cross-functional MOM teams to streamline data-collection workflows and ensure consistency.
- Synthesized findings into concise reports and presentations for labor-market research and program evaluation.

AI & Data Analytics Consultant, (Internship)

07/2022 - 02/2023

Deloitte Consulting

Supported consulting engagements with end-to-end data analytics and machine learning solutions, from proposal development through model deployment and stakeholder reporting.

Key Projects

- **Consulting Proposal Framework**
Translated client business requirements into analytical solution roadmaps and data-driven proposal artifacts.
- **Time-Series Forecasting Optimization**
Enhanced ARIMA-based models with feature engineering and evaluated performance improvements.
- **Predictive Modeling Pipelines**
Implemented gradient boosting solutions (XGBoost, LightGBM) within scikit-learn workflows to boost predictive accuracy.
- **ETL & Data Pipeline Automation**
Queried relational databases and ETL processes to automate data extraction, transformation, and loading for rapid model iteration.

Core Tasks

- Validated, cleansed, and structured large datasets using Python (pandas) to ensure integrity for analytics workflows.
- Developed SQL queries and Python scripts to streamline data retrieval from multiple sources.
- Built and tuned machine-learning models, conducting hyperparameter searches and cross-validation.
- Collaborated with consulting teams to translate analytical findings into actionable insights and client deliverables.
- Prepared and delivered presentations of technical results, aligning recommendations with client objectives.

Technical Logistics Coordinator (Contract)

07/2021 - 09/2021

Hitachi

Supported technical operations by inspecting server hardware, managing inventory records, and ensuring seamless equipment staging for IT deployments.

Key Project

- Hardware Staging Optimization**
Collaborated with IT and engineering teams to redesign the server staging workflow, reducing setup bottlenecks and improving readiness for rollouts.

Core Tasks

- Inspected and assembled high-performance servers and computing components, upholding stringent hardware QA standards.
- Maintained accurate inventory and supply records in ERP and custom supply-chain databases.
- Conducted root-cause analyses on shipment discrepancies to enable proactive restocking and minimize workflow interruptions.
- Partnered with cross-functional teams to coordinate hardware staging, ensuring timely delivery and deployment.

LEADERSHIP & ACTIVITIES

- Vice Chairperson**, Officer Cadet Course
- 19-day Expeditionary SPF Leadership Development Course**, Outward Bound
- Squad Person In-Charge**, Police Officer Basic Course
- Class reasurer**, Polytechnic
- Subcommittee Infocommunication echnology Study Club (IITSC)**, Polytechnic
- Executive Student Leader**, Secondary School
- STEM Club, Badminton Club**, Secondary School
- Class Monitor**, Primary School
- Art Club**, Primary School

HONORS & AWARDS

- Best Operational Fitness Award**, Officer Cadet Course2024
- Best Trainee Award**, Police Officer Basic Course2023
- Best Operational Fitness Award**, Police Officer Basic Course2023
- Edusave Good Progress Award**2023
- MUIS Madrasah Award Secular Subjects**2017, 2020
- Edusave Certificate Of Academic Achievement**2015

CERTIFICATIONS

- RPA Developers Training Course (Uipath Studio)**, simplifynext02/2025
- Senior Officer Course**, Singapore Police Force11/2024
- Police Basic Course**, Singapore Police Force01/2024
- CS50's, Harvard University's introduction to Computer Science**, edX03/2023
- IMP .AC.T Externship**, Great Eastern Financial Advisers Pte Ltd01/2023
- Python for Data Science and Machine Learning Bootcamp**, Udemy01/2023
- RPA Developer Foundation**, UiPath04/2022
- HTML Essential T raining**, LinkedIn,03/2022
- Python Data Structures & Algorithms**, LinkedIn03/2022
- Food Hygiene Certificate**, Singapore Food Agency11/2019

SKILLS

Data & Analytics

Python (pandas, NumPy), SQL (PostgreSQL, BigQuery), Data Cleaning & Preprocessing, Exploratory Data Analysis (EDA), ETL Pipelines, Time-Series Forecasting, Feature Engineering, Predictive Modeling, Model Evaluation, Cross-Validation, Scikit-learn, XGBoost, LightGBM

Data Engineering & Systems

RESTful APIs, Backend Services, Real-Time & Near-Real-Time Data Processing, Data Pipelines, Automation (UiPath), Cloud Platforms (Google Cloud Platform), System Integration, Data Validation & Quality Assurance, Security & Governance-Aware Systems

Visualization & Decision Support

Power BI, Tableau, ArcGIS, Geospatial Analytics, Interactive Dashboards, Stakeholder Reporting, Decision-Support Systems

Software & Tools

Git (Version Control), Python Scripting, Excel (Advanced Formulas, VBA), KNIME, WebXR, Babylon.js, Linux/Unix Environments

Professional & Transferable Skills

Agile / Sprint-Based Development, Cross-Functional Collaboration, Technical Documentation, Requirements Translation, Structured Problem-Solving, Clear Technical Communication, Adaptability in High-Pressure Environments